

CHEATSHEET · QUICK REFERENCE

Lab 8 — IP, VLAN, DNS, DHCP, FQDN, and Hosts File

Configure a static IP and VLAN, memorise RFC 1918 ranges, stand up a BIND9 zone and an ISC DHCP scope, and see how APIPA and /etc/hosts behave.

IP, gateway and VLAN

```
ip addr add 10.99.99.10/24 dev eth0
Assign a static IP

ip route add default via 10.99.99.1 dev eth0
Set the default gateway

ip link add link eth0 name eth0.20 type vlan id 20
802.1Q VLAN sub-interface

ip addr add 192.168.20.10/24 dev eth0.20
Address the VLAN interface
```

RFC 1918 private ranges (memorise)

Prefix	Range	Mask
10.0.0.0/8	10.0.0.0 - 10.255.255.255	/8
172.16.0.0/12	172.16.0.0 - 172.31.255.255	/12
192.168.0.0/16	192.168.0.0 - 192.168.255.255	/16
169.254.0.0/16	APIPA (DHCP failed)	/16

DNS, hosts and DHCP

```
dig @127.0.0.1 web.lab.local +short
Resolve an A record from BIND9

echo "10.99.99.50 app.lab.local app" >> /etc/hosts
Local override that beats DNS

getent hosts app.lab.local
Resolve via nsswitch (hosts first)

dhcpd -t -cf /etc/dhcp/dhcpd.conf
Validate a DHCP scope config

ip addr add 169.254.42.42/16 dev eth0
Simulate an APIPA address
```